








Advanced Time Series Analysis and Forecasting

Chapter 2: Structural breaks and nonlinear models



Daniel Traian PELE

Bucharest University of Economic Studies
IDA Institute Digital Assets
Blockchain Research Center
AI4EFin Artificial Intelligence for Energy Finance
Romanian Academy, Institute for Economic Forecasting
MSCA Digital Finance

Acronyms used in this material (1/3)

- **ADF**: Augmented Dickey–Fuller (test)
- **AI**: Artificial Intelligence
- **AIC**: Akaike Information Criterion
- **AR**: AutoRegressive (model); in event studies: Abnormal Return
- **ARCH**: AutoRegressive Conditional Heteroskedasticity
- **ARDL**: AutoRegressive Distributed Lag (model)
- **BDS**: Brock–Dechert–Scheinkman (test)
- **BIC**: Bayesian Information Criterion
- **BLS**: Bureau of Labor Statistics (US)
- **BNR**: Banca Națională a României (the National Bank of Romania)
- **CI**: Confidence Interval
- **COVID-19**: COronaVIrus Disease 2019
- **CPI**: Consumer Price Index
- **CRPS**: Continuous Ranked Probability Score
- **CSW**: Chu, Stinchcombe and White (monitoring boundary)
- **CUSUM**: CUmulative SUM
- **DM**: Diebold–Mariano (test)
- **ESTAR**: Exponential Smooth Transition AutoRegression
- **FCLT**: Functional Central Limit Theorem
- **FRED**: Federal Reserve Economic Data
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity

Acronyms used in this material (2/3)

- **GDP**: Gross Domestic Product
- **GW**: Giacomini–White (test of conditional predictive ability)
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **HICP**: Harmonised Index of Consumer Prices
- **HLN**: Harvey–Leybourne–Newbold (small-sample correction of the DM test)
- **ICSS**: Iterated Cumulative Sums of Squares
- **IFS**: International Financial Statistics (IMF)
- **IMF**: International Monetary Fund
- **JAE**: Journal of Applied Econometrics
- **KSS**: Kapetanios, Shin and Snell (unit-root test against ESTAR)
- **LLM**: Large Language Model
- **LM**: Lagrange Multiplier
- **LM1**: Lagrange Multiplier linearity test, first-order expansion
- **LM3**: Lagrange Multiplier linearity test, third-order expansion
- **LR**: Likelihood Ratio (test)
- **LS**: Least Squares
- **LSTAR**: Logistic Smooth Transition AutoRegression
- **LWZ**: Liu, Wu and Zidek (modified Schwarz criterion)
- **MFM**: Modelling Financial Markets

Acronyms used in this material (3/3)

- **MOSUM**: MOving SUM (of residuals)
- **MPQ**: McConnell and Perez-Quiros (2000)
- **MSFE**: Mean Squared Forecast Error
- **OECD**: Organisation for Economic Co-operation and Development
- **OLS**: Ordinary Least Squares
- **OLS-CUSUM**: CUSUM of OLS residuals
- **ONS**: Office for National Statistics (UK)
- **PELT**: Pruned Exact Linear Time (change-point algorithm)
- **PPP**: Purchasing Power Parity
- **RMSE**: Root Mean Squared Error
- **ROBOR**: Romanian Interbank Offer Rate
- **SCA**: Seasonally and Calendar Adjusted
- **SETAR**: Self-Exciting Threshold AutoRegression
- **SSR**: Sum of Squared Residuals
- **STAR**: Smooth Transition AutoRegression
- **SVAR**: Structural Vector AutoRegression
- **TAR**: Threshold AutoRegression
- **TSA**: Time Series Analysis
- **UK**: United Kingdom
- **US**: United States
- **VAR**: Vector AutoRegression
- **VECM**: Vector Error Correction Model

Today's question and route

- **Question**: when do the parameters of a time-series model change, and when is the dynamics itself state dependent?
  - a **break** is a change in the parameters at some dates; a **nonlinear** model lets the parameters depend on the past of the series
  - both produce the same symptoms: unstable forecasts, spurious persistence, rejected linearity
- **Route** of the chapter
  - breaks: Chow, sup-Wald (Andrews), Bai–Perron, monitoring (CUSUM, CSW), variance breaks (ICSS), unit roots with breaks
  - forecasting under breaks: the choice of the estimation window
  - nonlinear models: TAR/SETAR, STAR (LSTAR, ESTAR), nonlinearity tests, nonlinear forecasts
- We build on TSA, Chapter 3 (Perron (1989), Zivot and Andrews (1992)) and on Chapters 0 and 1 (HAC, bootstrap, DM tests); Seminar 2 comes before this lecture

Learning outcomes

- Derive the Chow test and explain why searching over the break date changes its null distribution (the Davies problem)
- Test for one or several breaks with unknown dates (sup-, exp-, ave-Wald, Bai–Perron), select their number and build confidence intervals for the dates
- Monitor a model in real time with a controlled false-alarm rate, and detect breaks in variance
- Choose the estimation window when the data contain breaks, and evaluate the choice out of sample
- Specify, test, estimate and forecast threshold and smooth-transition models, with valid inference when the threshold is not identified under the null

Reading, data and tools

- Surveys: Casini and Perron (2019), Hansen (2001), Rossi (2013), van Dijk, Teräsvirta and Franses (2002), Hansen (2011)
  - textbooks: Hamilton (1994), Ch. 22 (regime changes); Kilian and Lütkepohl (2017); the forecasting survey Petropoulos et al. (2022) (open access); bridge: Huang and Petukhina (2022)
- Python Quantlets of this chapter: Quantlets/Ch_02
  - Bai–Perron dynamic programming, sup-Wald, ICSS, CSW monitoring, Hansen's bootstrap and LR interval, LSTAR and ESTAR by nonlinear least squares, all written out in `numpy/scipy`
  - R users: the `strucchange` package Zeileis et al. (2002) implements the same break tests
- Lecture notebook: open in Google Colab

Data used in this chapter

Series	Source	Use
US ex-post real interest rate, 1961–1986	JAE data archive (Bai–Perron); rebuilt from FRED (TB3MS, CPIAUCNS)	Bai–Perron
US real GDP growth, quarterly	FRED (GDPC1)	variance break
HICP inflation, Romania (y/y and m/m)	Eurostat (prc_hicp_minr)	breaks, monitoring
EUR/RON, daily	BNR reference rate	ICSS
Real GDP, Romania, quarterly	Eurostat (namq_10_gdp)	unit roots
Unemployment rate, US men aged 20+	BLS via FRED (LNS13000025 / LNS11000025; not adjusted: LNU03000025 / LNU01000025)	TAR, LSTAR
Real dollar-sterling rate	FRED (DEXUSUK, CPIAUCNS), OECD and ONS UK CPI	ESTAR
Canadian lynx trappings, 1821–1934	R data set read by statsmodels	SETAR

- All sources are public and need no account or key; monthly data up to August–September 2026
- HICP: harmonised index of consumer prices; BLS: US Bureau of Labor Statistics; ONS: UK Office for National Statistics; JAE: Journal of Applied Econometrics

Origins of break tests and threshold models



Gregory C. Chow

Portrait: unknown author (1950s–1960s); public domain; Wikimedia Commons



Howell Tong

Photo: Mr and Mrs Howell Tong / IMS; CC BY 3.0; Wikimedia Commons

- 1960: Chow (1960) tests whether two regressions share their coefficients; the same year Quandt (1960) maximises the likelihood ratio over the switch date
- 1980: Tong and Lim (1980) propose threshold autoregression: a linear model in each regime, nonlinear as a whole, able to produce limit cycles

Setting and notation

- Linear model $y_t = x_t'\beta_t + u_t$, $t = 1, \ldots, T$; stability means H_0: $\beta_t = \beta$ for all t
  - x_t: the $k \times 1$ vector of regressors; β_t: the coefficients at date t; u_t: the error
  - one break at $T_1 = [\pi T]$ ($[\cdot]$: the integer part): $\beta_t = \beta_1$ for $t \leq T_1$ and $\beta_t = \beta_2$ after; $\pi \in (0, 1)$ is the **break fraction**
  - m breaks $T_1 < \cdots < T_m$ split the sample into $m + 1$ regimes
- Kinds of change
  - **pure** change (all coefficients) or **partial** change ($y_t = x_t'\beta + z_t'\delta_j + u_t$: only δ moves)
  - z_t: the regressors whose coefficients change; δ_j: their value in regime j; β: the coefficients common to all regimes
  - in the mean, in the dynamics, in the variance; abrupt or gradual (random-walk coefficients, Stock and Watson (1996))
- In TSA, Chapter 3 a break was a nuisance for unit-root tests; here the break is the object of inference

The Chow test (1/2)

- Known date T_1, k regressors, i.i.d. Normal errors Chow (1960)
  - SSR: the sum of squared residuals; S_0: of the pooled regression on all T observations; S_1, S_2: of the regressions before and after T_1
- The test: how much the fit improves when each sub-sample has its own coefficients
  - $F = \dfrac{(S_0 - S_1 - S_2)/k}{(S_1 + S_2)/(T - 2k)} \sim F(k, T - 2k)$ under H_0
  - $S_0 - S_1 - S_2 \geq 0$: the reduction of the SSR, per extra coefficient; a large F is evidence of a break
  - equivalent to the F test of $\delta = 0$ in $y_t = x_t'\beta + (x_t 1\{t > T_1\})'\delta + u_t$; δ: the change of the coefficients after T_1

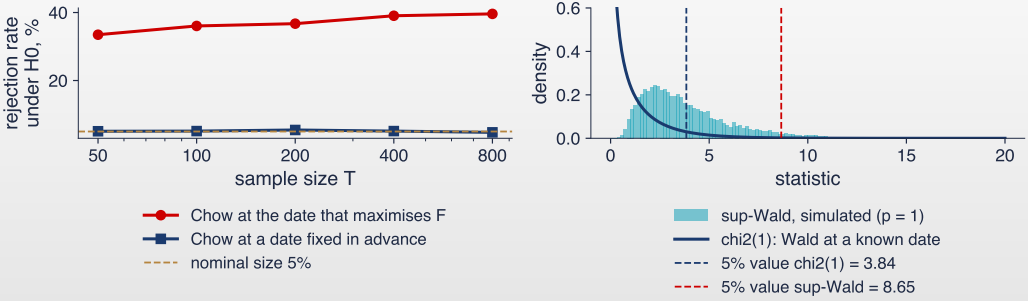
The Chow test (2/2)

- Robust version: the Wald statistic $W_T(T_1) = \hat{\delta}' \hat{V}_{\delta}^{-1} \hat{\delta}$, asymptotically $\chi^2(k)$
  - $\hat{V}_{\delta}$: a heteroskedasticity- or HAC-robust covariance matrix of $\hat{\delta}$ (Chapter 0)
  - the classical F also assumes equal variances in the two regimes
- **Predictive** Chow test when $T - T_1 < k$ (too few observations after the break to estimate the model):
  - $F = \dfrac{(S_0 - S_1)/(T - T_1)}{S_1/(T_1 - k)}$
  - it asks whether the model estimated up to T_1 forecasts the last observations; the ancestor of forecast-breakdown tests (Section 7)

A break date chosen after inspecting the data

- In practice T_1 is rarely known: the researcher sees a jump in the plot and tests there
  - the test then uses the date that *maximises* the statistic: a data-snooping problem (Chapter 0)
- Under H_0 the break date is **not identified**: the parameter π appears only under the alternative Davies (1987)
  - standard asymptotics fail; the null distribution of $\sup_\pi W_T(\pi)$ is that of the supremum of a stochastic process
- The same problem returns for the threshold of a TAR model (Section 8) and for the transition parameter of a STAR model (Section 9)

A Chow test at the data-chosen date



⊡ Left: white noise, mean-shift Chow test at 5%, 4 000 replications per T; right: simulated null density of $\sup_\pi W_T(\pi)$, $\pi \in [0.15, 0.85]$, against $\chi^2(1)$

ATS_ch2_chow_snooping

Interpreting the size distortion

- At a fixed date the test keeps its size (5.1% for $T = 100$); at the maximising date it rejects a true null in 33% to 40% of samples
  - the distortion grows slowly with T: more candidate dates, a larger maximum
- The 5% critical value of the supremum is 8.65, against 3.84 for $\chi^2(1)$; with two coefficients 11.69
- Lesson: the date search is part of the test; the critical values must account for it

Recap: Instability and known dates

- A break is a change of β_t; pure or partial, in mean, dynamics or variance
- Chow: an F test valid only for a date fixed before seeing the data
- Choosing the date from the data multiplies the size several times
- The break date is a nuisance parameter not identified under H_0

The sup-Wald test of Andrews (1993) (1/2)

- Compute the Chow–Wald statistic $W_T(\pi)$ at every candidate break fraction and take the largest Andrews (1993)
  - $\sup_{\pi \in \Pi} W_T(\pi), \quad \Pi = [\pi_0, 1 - \pi_0]$
  - Π: the candidate fractions; π_0: the **trimming**, usually 0.15, so that each regime has at least 15% of the sample
  - LM, Wald and LR versions share the same limit; Quandt (1960) used the LR version
- **Theorem** Andrews (1993): under H_0 and regularity conditions (stationary, mixing regressors)
  - $W_T(\cdot) \Rightarrow Q_p(\pi) = \dfrac{\|B_p(\pi) - \pi B_p(1)\|^2}{\pi(1-\pi)}$
  - $\Rightarrow$: convergence of the whole process in π; B_p: a p-dimensional standard Brownian motion; p: the number of coefficients allowed to break
  - $B_p(\pi) - \pi B_p(1)$ is a Brownian bridge; $\pi(1-\pi)$ is its variance: $Q_p(\pi)$ is a squared, standardised bridge (▸ Appendix)

The sup-Wald test of Andrews (1993) (2/2)

- Limit: $\sup W_T \to_d \sup_{\pi \in \Pi} Q_p(\pi)$
  - at a fixed π, $Q_p(\pi) \sim \chi^2(p)$; the supremum over many π is much larger
- Why trimming: $\sup_{\pi \in (0,1)} Q_p(\pi) = \infty$ almost surely (law of the iterated logarithm)
  - critical values depend on p and π_0; here simulated: $p = 1$: 7.11 (10%), 8.65 (5%), 12.17 (1%)

Optimal tests: Andrews and Ploberger (1994)

- The supremum is not the only functional; averaging is optimal against alternatives close to H_0 Andrews and Ploberger (1994)
  - $|\Pi| = 1 - 2\pi_0$: the length of the set of candidate fractions; both statistics average over it
  - $\mathrm{expW} = \ln\left(\frac{1}{|\Pi|}\int_{\Pi} \exp\left(\frac{1}{2}W_T(\pi)\right)d\pi\right)$: weighted average power, for medium and large breaks
  - $\mathrm{aveW} = \frac{1}{|\Pi|}\int_{\Pi} W_T(\pi)\,d\pi$: optimal for very small breaks, close to the Nyblom test of random-walk coefficients
- 5% values for $p = 1$, $\pi_0 = 0.15$ (simulated): exp 2.02, ave 2.87; sup: previous slide
- The sup-Wald test has power against *any* kind of instability, not only a single break
  - a rejection says "unstable", not "one break at $\hat{\pi}$"; the number of breaks is the subject of Section 3

Estimating the break date (1/2)

- Least squares: choose the date that minimises the total SSR of the two regimes
  - $\hat{T}_1 = \arg\min_{T_1} [S_1(T_1) + S_2(T_1)]$
  - $S_1(T_1)$, $S_2(T_1)$: the SSR before and after a candidate date T_1; equivalently the date of the largest Wald statistic under homoskedasticity
- Precision Bai (1997)
  - for a fixed break size, $\hat{T}_1 - T_1 = O_p(1)$: the **date** error is bounded in probability and does not grow with T
  - so the fraction $\hat{\pi} = \hat{T}_1 / T$ converges at rate T, faster than the usual $\sqrt{T}$
  - the regime coefficients are $\sqrt{T}$-consistent and asymptotically Normal as if the date were known

Estimating the break date (2/2)

- Confidence interval for the date Bai (1997): with a shrinking break $\Delta_T \to 0$
  - $\Delta_T^2 \sigma^{-2}(\hat{T}_1 - T_1) \to_d \arg\max_s Z(s), \quad Z(s) = W(s) - |s|/2$
  - Δ_T: the size of the change in the mean; σ^2: the error variance; $W(s)$: a two-sided Brownian motion, $s \in \mathbb{R}$
  - the interval is the date $\pm$ quantiles of $\arg\max Z$, rescaled by $\sigma^2/\hat{\Delta}^2$
- Reading
  - large breaks, small noise: a short interval; the interval is asymmetric when variances differ across regimes

Case study: McConnell and Perez-Quiros (2000)

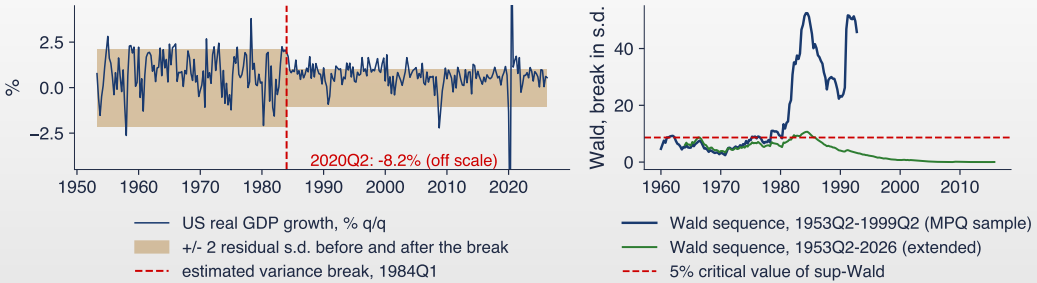


Gasoline queue, Maryland, 15 June 1979

Photo: Warren K. Leffler (1979), Library of Congress; public domain; Wikimedia Commons

- The paper: US real GDP growth, 1953Q2–1999Q2; AR(1) for the conditional mean; then a break in the mean of $\sqrt{\pi/2}\,|\hat{e}_t|$, an unbiased estimate of the standard deviation under Normality McConnell and Perez-Quiros (2000)
  - $\hat{e}_t$: the AR(1) residual; here $\pi = 3.14\ldots$; if $e_t \sim N(0, \sigma^2)$, $\mathbb{E}|e_t| = \sigma\sqrt{2/\pi}$, hence the factor
  - tests: sup-, exp-, ave-Wald Andrews (1993), Andrews and Ploberger (1994); their result: a variance break in 1984Q1, the "Great Moderation"
- Our replication: the same two steps on FRED GDPC1 (today's vintage), robust Wald statistics, simulated p-values; then the sample extended to 2026
- Question: does the variance break survive revised data and the COVID-19 quarters?

The Great Moderation in US output growth



- Top: growth with ± 2 residual standard deviations before and after the estimated break; bottom: the Wald sequence for a break in $\mathbb{E}\sqrt{\pi/2}|e_t|$ (HAC), two samples

ATS_ch2_great_moderation

Interpreting the variance break

- MPQ sample ($T = 185$): sup-Wald 52.5 ($p < 0.001$), exp 23.2, ave 17.5; date 1984Q1, the date found by McConnell and Perez-Quiros (2000)
  - residual s.d. 1.06 before, 0.45 after: the variance falls by a factor of 5.5; no break in the mean equation (sup 3.2, p 0.861)
- To 2019: the break moves by one quarter (1984Q2), variance ratio 4.1; the Great Moderation survived the 2008–2009 recession
- To 2026: sup-Wald falls to 10.7 (p 0.019) and the ratio to 1.1: two outliers (2020Q2: -8.2%) dominate the post-break variance
  - a second break, or outliers? A one-break test cannot say; robust scale estimators or a second break are needed

Recap: One break at an unknown date

- sup-Wald: the supremum of a squared Brownian bridge; trim, and use its own critical values
- exp- and ave-Wald are optimal for moderate and small breaks
- The date is estimated precisely, the CI comes from $\arg\max_s \{W(s) - |s|/2\}$
- Variance breaks: test the mean of $|\hat{e}_t|$; outliers can mask or mimic them

The multiple-break model

- $y_t = x_t'\beta + z_t'\delta_j + u_t$, $t = T_{j-1}+1, \ldots, T_j$, $j = 1, \ldots, m+1$, with $T_0 = 0$, $T_{m+1} = T$ Bai and Perron (1998)
  - $T_1 < \cdots < T_m$: the break dates; δ_j: the coefficients of z_t in regime j; β: those of x_t, common to all regimes
  - pure change: no x_t; mean-shift model: $z_t = 1$
  - minimum segment length $h = [\varepsilon T]$, usually $\varepsilon = 0.15$; errors may be autocorrelated and heteroskedastic across regimes
- Estimator: the **global** minimiser of the SSR over all admissible partitions $(T_1, \ldots, T_m)$
  - break fractions consistent at rate T, as for one break; the dates are estimated jointly, not one at a time
- Brute force needs $O(T^m)$ regressions; dynamic programming needs $O(T^2)$ Bai and Perron (2003)

Dynamic programming

- Step 1: compute $\mathrm{SSR}(i, j)$ of one regression on every segment $[i, j]$ with $j - i + 1 \geq h$ (recursive residuals: $O(T^2)$ operations)
- Step 2: Bellman recursion for the best r-break partition of the first n observations
  - $\mathrm{SSR}(\{T_{r,n}\}) = \min_{rh \leq j \leq n-h} \left[\mathrm{SSR}(\{T_{r-1,j}\}) + \mathrm{SSR}(j+1, n)\right]$
  - $\{T_{r,n}\}$: the best partition of observations $1, \ldots, n$ with r breaks; j: the date of its last break; h: the minimum segment length
  - the optimal m-break partition contains optimal partitions of its prefixes (Bellman's principle)
- One pass gives the optimal partitions for every $m \leq M$: the input of all tests and criteria
  - partial change (β common) needs an iteration between β and the partition Bai and Perron (2003); Seminar 2, A3 runs the recursion by hand

Testing and selecting the number of breaks (1/2)

- $\sup F_T(k)$: no break against exactly k breaks, at the SSR-optimal partition, with robust variance Bai and Perron (1998)
  - M: the largest number of breaks considered (here 5)
  - **UDmax** $= \max_{k \le M} \sup F_T(k)$: no break against an unknown number of breaks, up to M
  - **WDmax**: the same, with each k weighted by the ratio of critical values
- **Sequential** $\sup F_T(\ell + 1 \mid \ell)$: ℓ breaks against $\ell + 1$
  - for each of the $\ell + 1$ segments, the largest single-break statistic; add a break while it rejects
  - recommended strategy Bai and Perron (2003): UDmax or WDmax first, then the sequential tests

Testing and selecting the number of breaks (2/2)

- Information criteria: pick the m with the smallest value
  - $\mathrm{BIC}(m) = \ln(\mathrm{SSR}_m/T) + p^* \ln T/T$
  - **LWZ**: $\ln \frac{\mathrm{SSR}_m}{T-p^*} + p^* \frac{0.299}{T} (\ln T)^{2.1}$ Liu, Wu and Zidek (1997)
  - SSR_m: the SSR of the best m-break partition; p^*: the number of coefficients and break dates; the second term penalises extra breaks
- Known weaknesses
  - BIC picks too many breaks under serial correlation, LWZ too few with small breaks

Case study: Bai and Perron (2003), the US real interest rate

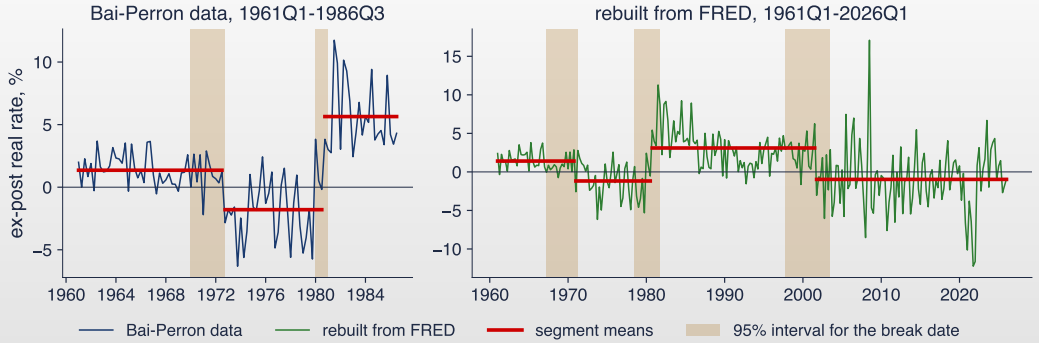


Paul Volcker (left) and Ronald Reagan, December 1981

Photo: White House Photographic Office (1981); public domain; Wikimedia Commons

- The paper, Section 6.1: the ex-post real rate of Garcia and Perron (1996), quarterly 1961Q1–1986Q3 ($T = 103$), mean-shift model, $\varepsilon = 0.15$ ($h = 15$), $M = 5$, HAC variances that differ across segments Bai and Perron (2003)
  - our replication uses their data file (JAE data archive) and the same settings
- Extension: the series rebuilt from FRED (T-bill rate at the end of the quarter minus next-quarter CPI inflation), correlation 0.99 with the original, 1961Q1–2026Q1
- Question: how many regimes, and do they match the history of monetary policy?

Mean shifts in the US real interest rate



- Red: segment means of the BIC/LWZ partition; shaded: 95% intervals for the dates Bai (1997), heterogeneous long-run variances. Top: original data; bottom: rebuilt to 2026

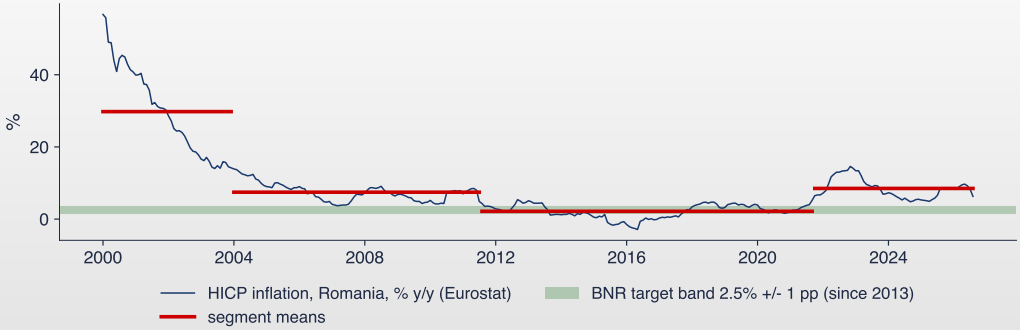
ATS_ch2_bai_perron

Interpreting the Bai–Perron results

m	0	1	2	3	4	5
SSR	1215	645	456	445	–	–
$\sup F_T(m)$	–	61.1	46.1	36.1	27.4	20.6
5% value	–	8.74	7.27	6.05	–	–
BIC	2.513	1.970	1.713	1.779	–	–
LWZ	2.550	2.082	1.901	2.043	–	–

- UDmax 61.1 (5%: 8.87); sequential: $F(2|1) = 36.9$, $F(3|2) = 16.1$, $F(4|3) = 0.04$ (5%: 10.54, 11.56): 3 breaks, 1966Q4, 1972Q3, 1980Q3
  - BIC and LWZ choose 2 breaks: 1972Q3, 1980Q3; the dates are the ones reported by Bai and Perron (2003)
- Means 1.36%, -1.80%, 5.64%: negative real rates in the 1970s, then the 1979–1981 disinflation; 95% intervals [1970Q1, 1972Q4], [1980Q1, 1981Q1]
- Extended sample: 3 breaks (1970Q4, 1980Q3, 2001Q3); the last regime has a mean of -0.97%: the low-rate era after 2001, interval [1997Q4, 2003Q3]

Mean shifts in Romanian inflation



- HICP inflation, y/y, January 2000–August 2026 ($T = 320$), Bai–Perron mean shifts, $\varepsilon = 0.15$, $M = 5$; red: the partition chosen by BIC and LWZ

ATS_ch2_ro_inflation_breaks

Interpreting the Romanian inflation regimes

- BIC and LWZ: 3 breaks (December 2003, July 2011, September 2021); means 29.8%, 7.5%, 2.2%, 8.5%
  - disinflation, inflation targeting (adopted in August 2005), the low-inflation decade, the 2021–2023 surge
- The robust sequential procedure stops at 1 break: $\sup F(1) = 12.8$, but $F(2|1) = 2.06$
  - inflation is very persistent within regimes: the HAC variance absorbs the shifts, and the test loses power
  - for the same reason the 95% intervals for the dates span years (second date: [January 2000, December 2014]); they are not drawn
- Better model: partial change in the intercept of an AR model, so that the errors are close to white noise (project idea)

Change-point detection in statistics and machine learning

- Penalised segmentation: minimise $\sum_j \mathrm{cost}(\text{segment } j) + \beta\, m$ over partitions
  - cost: e.g. the SSR or minus the log-likelihood of a segment; $\beta > 0$: the penalty per break; m: the number of breaks
  - PELT prunes the Bellman recursion to $O(T)$ under mild conditions Killick, Fearnhead and Eckley (2012); binary segmentation and wild binary segmentation Fryzlewicz (2014) scale to long series
  - the Python package `ruptures` implements them Truong, Oudre and Vayatis (2020); the BIC of Bai–Perron is a particular penalty
- What econometrics adds: inference on the number of breaks and confidence intervals for dates under serial correlation
  - common breaks in systems of equations Oka and Perron (2018); survey Casini and Perron (2019)
- Engineering practice tunes the penalty on labelled data; econometrics fixes the size of a test: both are valid if the choice is pre-registered

Recap: Multiple breaks

- Global least squares over all partitions, by dynamic programming in $O(T^2)$
- UDmax/WDmax, then sequential $F(\ell+1|\ell)$; BIC and LWZ as a cross-check
- US real rate: breaks in 1972Q3 and 1980Q3 replicated; a further regime after 2001
- Persistent errors: robust tests lose power and date intervals become wide

CUSUM and MOSUM tests (1/2)

- Recursive residuals Brown, Durbin and Evans (1975): the standardised one-step forecast error of the model estimated up to $t-1$
  - $w_t = (y_t - x_t'\hat{\beta}_{t-1})/\sqrt{1 + x_t'(X_{t-1}'X_{t-1})^{-1}x_t}$
  - $\hat{\beta}_{t-1}$: OLS on observations $1, \ldots, t-1$; X_{t-1}: the matrix of their regressors; the root rescales for estimation uncertainty
  - under H_0 with Normal errors, w_t are i.i.d. $N(0, \sigma^2)$
- CUSUM: the cumulated standardised recursive residuals
  - $W_r = \hat{\sigma}^{-1} \sum_{t=k+1}^{r} w_t$, 5% boundary $\pm 0.948[\sqrt{T-k} + 2(r-k)/\sqrt{T-k}]$
  - k: the number of regressors (the first k observations start the recursion); W_r behaves like a Brownian motion; a crossing signals a break

CUSUM and MOSUM tests (2/2)

- What CUSUM detects
  - changes in the intercept and a systematic drift of forecast errors; weak against changes that average out
- OLS-CUSUM uses ordinary residuals: a Brownian bridge limit Ploberger and Krämer (1992); MOSUM sums over a moving window of fixed width Chu, Hornik and Kuan (1995)
  - MOSUM reacts faster to a break in the middle of the sample and to temporary changes
- These are **retrospective** tests: the whole sample is available when the test is run

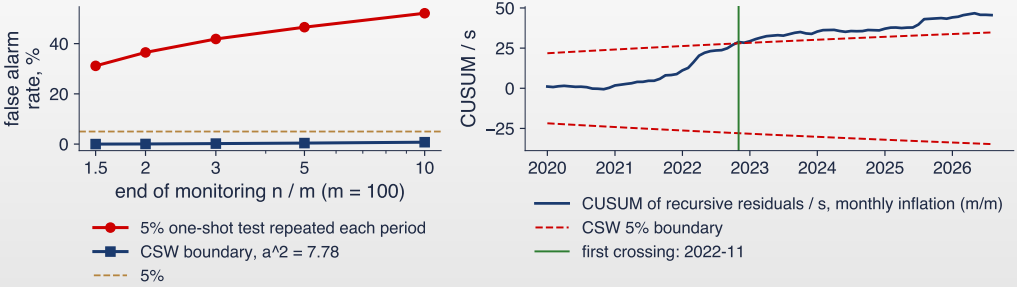
Monitoring: the failure of repeated tests (1/2)

- A central bank estimates a model on m historical observations and asks, at every new release, "has it broken down?"
  - a 5% test repeated at $n = m+1, m+2, \ldots$ rejects a true H_0 eventually with probability 1: the law of the iterated logarithm
- Chu, Stinchcombe and White (1996): monitor the CUSUM of the new recursive residuals and stop the first time it crosses a widening boundary
  - $Q_n = \hat{\sigma}^{-1} \sum_{t=m+1}^{n} w_t$, alarm if $|Q_n| > \sqrt{n\,[a^2 + \ln(n/m)]}$
  - n: the current sample size; a: a constant chosen to fix the false-alarm probability; the $\ln(n/m)$ term widens the boundary as time passes

Monitoring: the failure of repeated tests (2/2)

- The boundary comes from Robbins and Siegmund (1970):
  - $P\{\exists u \geq 1 : |W(u) - W(1)| \geq \sqrt{u(a^2 + \ln u)}\} = 2[1 - \Phi(a) + a\varphi(a)]$
  - W: a standard Brownian motion; $u = n/m$; Φ, φ: the standard Normal distribution and density functions
  - $a^2 = 7.78$: size 5.1% over an *infinite* horizon; $a^2 = 6.25$: 10.0%
- Forecast-based monitoring: forecast breakdowns Giacomini and Rossi (2009) and the fluctuation test of relative accuracy Giacomini and Rossi (2010) (Chapter 1)

False alarms and real-time monitoring of Romanian inflation



⊡ Left: white noise, $m = 100$, 2 000 replications; right: mean of monthly HICP inflation (m/m), historical sample 2015–2019, monitoring from January 2020

ATS_ch2_monitoring

Interpreting the monitoring results

- Repeated 5% tests raise a false alarm in 36% of samples by $n = 2m$ and in 52% by $n = 10m$; the CSW boundary in 0.8%
  - the CSW size is reached only as $n \to \infty$: over a finite horizon the boundary is conservative
- Romania: historical mean 0.15% per month, $\hat{\sigma} = 0.51$; the CUSUM crosses the boundary in November 2022
  - the repeated naive test would have signalled in January 2022; with a long-run $\hat{\sigma}$ (HAC) the signal comes only in August 2025
- The price of a controlled false-alarm rate is delay: nearly two years after inflation started to rise in 2021

Recap: Monitoring

- CUSUM and MOSUM of recursive residuals: retrospective fluctuation tests
- Repeating a one-shot test as data arrive guarantees a false alarm
- CSW boundary $\sqrt{n[a^2 + \ln(n/m)]}$ controls the size over an infinite horizon
- Romanian inflation: detection in late 2022, a long but honest delay

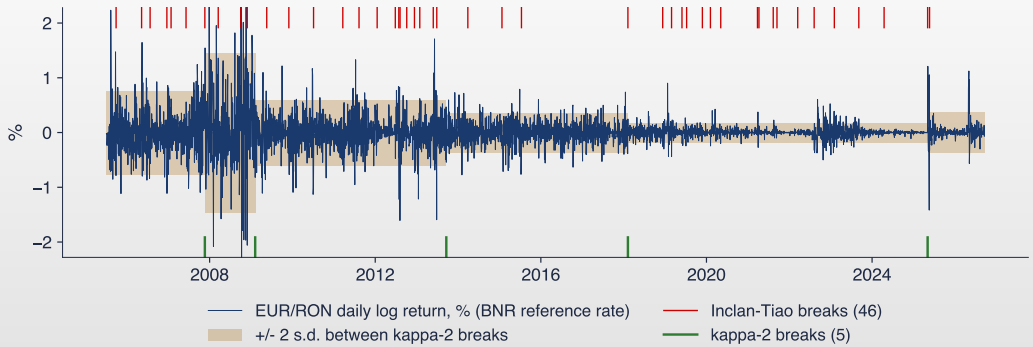
The ICSS algorithm and its correction (1/2)

- Inclán and Tiao (1994): compare the cumulated sum of squares with a straight line
  - $C_k = \sum_{t \le k} a_t^2, \quad D_k = C_k / C_T - k/T, \quad \mathrm{IT} = \sqrt{T/2}\max_k |D_k| \Rightarrow \sup |B^0(r)|$
  - a_t: the demeaned returns; C_k/C_T: the share of the total sum of squares reached at k; with a constant variance it grows like k/T
  - B^0: a Brownian bridge on $[0,1]$; 5% critical value 1.358; the date is the k of the largest $|D_k|$
- ICSS (iterated cumulative sums of squares)
  - apply the test iteratively to sub-segments (binary segmentation), then re-check each break between its neighbours

The ICSS algorithm and its correction (2/2)

- The factor $\sqrt{T/2}$ assumes $\mathrm{Var}(a_t^2) = 2\sigma^4$: i.i.d. Normal data
  - with kurtosis $\kappa = \mathbb{E}a_t^4/\sigma^4$ the statistic is inflated by $\sqrt{(\kappa-1)/2}$; with GARCH, a_t^2 is also autocorrelated
- κ_2 of Sansó, Aragó and Carrion (2004): the same CUSUM, scaled by a long-run variance
  - $\kappa_2 = \max_k |C_k - (k/T)C_T|/\sqrt{T\hat{\omega}_4}$
  - $\hat{\omega}_4$: the long-run variance of $a_t^2 - \hat{\sigma}^2$ (Bartlett kernel, Chapter 0)
  - same limit, valid under fat tails and conditional heteroskedasticity

Variance regimes of EUR/RON



- Daily log returns of the BNR reference rate, 5 352 days; red ticks: ICSS with the Inclán–Tiao statistic; green ticks and shaded bands: ICSS with κ_2

ATS_ch2_variance_breaks

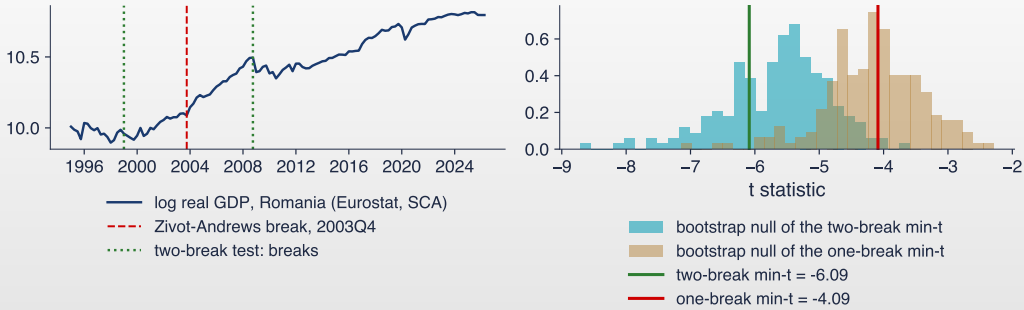
Interpreting the EUR/RON variance breaks

- Excess kurtosis 15.1: the Inclán–Tiao statistic is 24.0 and ICSS finds 46 breaks; $\kappa_2 = 4.69$ and 5 breaks
  - most Inclán–Tiao "breaks" are volatility clusters, not changes of the unconditional variance
- κ_2 dates: 20 November 2007, 6 February 2009, 19 September 2013, 7 February 2018, 5 May 2025
  - the global financial crisis, the end of its turbulence, the calmer managed float after 2018; daily s.d. between 0.09% and 0.73%
- For risk models (MFM, Chapter 5): a GARCH fitted across a variance break overstates persistence ($\alpha + \beta \to 1$) Diebold and Inoue (2001)

Extensions of the Zivot–Andrews test

- TSA, Chapter 3: a level shift mimics a unit root Perron (1989); Zivot and Andrews (1992) choose the date by minimising the ADF t statistic
  - their null is a unit root *without* a break: a break under the null makes the test reject too often
- Two breaks: Lumsdaine and Papell (1997) minimise the t statistic over two dates; Lee and Strazicich (2003) propose an LM test whose null allows breaks
  - Kim and Perron (2009): pre-test the break, then use a test valid under both hypotheses
- Critical values depend on the model, the trimming and the number of breaks; a sieve bootstrap under the null avoids the tables

Romanian GDP: unit root against a broken trend



- Log real GDP (SCA), 1995Q1–2026Q2; one and two breaks in level and trend; null distributions from a sieve bootstrap (AR on Δy_t, 199 replications)

ATS_ch2_unit_root_breaks

Interpreting the unit-root tests with breaks

- ADF with trend: -2.79 (p 0.201); Zivot–Andrews: -4.09 at 2003Q4 (5%: -5.07; bootstrap p 0.523)
- Two breaks (1999Q1 and 2008Q4): min-$t = -6.09$, bootstrap 5% value -7.16, p 0.266
  - the more breaks we allow, the more negative the critical value: the apparent gain disappears
- With about 30 years of quarterly data a unit root cannot be rejected; the convergence story (fast growth, then a 2009 break) remains a hypothesis

Recap: Breaks in variance and unit roots

- ICSS needs the κ_2 correction for fat tails and GARCH
- EUR/RON: a handful of genuine variance regimes, not dozens
- Unit-root tests with breaks: allow breaks under the null or pre-test them; bootstrap the critical values

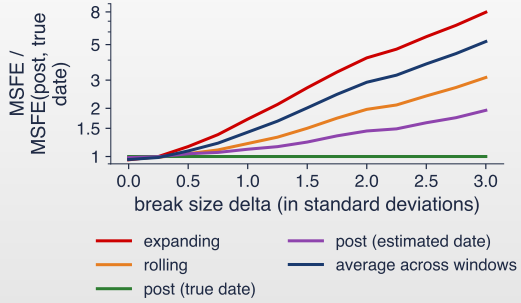
Location shifts and forecast failure

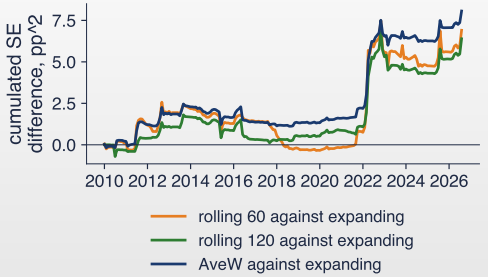
- Clements and Hendry (1998), Clements and Hendry (2006): in equilibrium-correction models, a shift of the equilibrium mean is the main source of systematic forecast errors
  - after the shift, the model keeps pulling the forecast towards the old mean
- Robust devices
  - **intercept correction**: add the last observed error to the forecast
  - **differencing**: forecast Δy; adapts after one period, at the cost of a larger variance
  - **shorter windows**: rolling estimation, or the post-break sample
- Survey of the evidence: Rossi (2013); window choice with time-varying parameters Inoue, Jin and Rossi (2017)

The choice of the estimation window

- Mean shift δ after n_1 pre-break observations, n_2 post-break; forecast with the last $w \geq n_2$ observations Pesaran and Timmermann (2007)
  - $\mathrm{MSFE}(w) = \sigma^2\left(1 + \frac{1}{w}\right) + \left(\frac{(w - n_2)\delta}{w}\right)^2$: variance falls with w, squared bias grows
  - MSFE: the mean squared forecast error of the sample mean over the window; σ^2: the noise variance; $(w - n_2)/w$: the share of pre-break data in the window
  - the optimal window includes *some* pre-break data when δ/σ is small or n_2 short (Seminar 2, A8)
- Feasible rules: estimate the date (Bai–Perron) and use the post-break sample; cross-validate w Pesaran and Timmermann (2007); average forecasts across windows (AveW) Pesaran and Pick (2011)
  - the date is estimated with error exactly when the break is small: post-break windows are then too short

Window choice: simulation and Romanian inflation





- Left: $T = 200$, break $n_2 = 20$ periods before the end, 4 000 replications, MSFE relative to the post-break mean with the true date; right: AR(2), one month ahead, cumulated squared-error gains over the expanding window

ATS_ch2_forecast_windows

Interpreting the window choice

- Simulation: without a break the expanding window wins (0.95); with $\delta = 1$: expanding 1.71, rolling 1.21, AveW 1.42, estimated post-break 1.11
  - $\delta = 2$: 4.13, 1.97, 2.91, 1.44: estimating the date pays when the break is large
- Romania, $P = 200$ months: RMSE expanding 0.664, rolling 60: 0.638, rolling 120: 0.640, AveW 0.633 pp
  - HLN against expanding: AveW 2.20 (p 0.029), rolling 60 1.19 (p 0.234); all gains arrive in 2022, the RMSE in 2021–2023 falls from 0.790 to 0.700
- Averaging across windows is a cheap insurance: never far from the best window, without estimating any date

Recap: Forecasting under breaks

- Location shifts drive forecast failure; intercept correction and differencing make forecasts robust
- Optimal window: a bias–variance trade-off that may include pre-break data
- AveW is robust when the break is uncertain; post-break windows when it is large and well dated
- Evaluate the window choice out of sample with DM–HLN Diebold and Mariano (1995), Harvey, Leybourne and Newbold (1997) (Chapter 1)

Motivation for nonlinear models



Canada lynx

Photo: Michael Zahra (2010); CC BY-SA 3.0; Wikimedia Commons

- Linear models are symmetric: a shock of -1 has the mirror effect of a shock of $+1$, at every state
  - business cycles: unemployment rises fast in recessions and falls slowly in expansions Neftçi (1984)
  - arbitrage: real exchange rates move freely inside a band of transaction costs, revert outside it Michael, Nobay and Peel (1997)
- Ecology: the 9–10 year cycle of the Canadian lynx is asymmetric (slow rise, fast fall); a linear AR cannot produce a stable cycle without noise
- A nonlinear model can: limit cycles, asymmetry, state-dependent persistence and impulse responses

Threshold and self-exciting threshold autoregression (1/2)

- Two-regime **TAR**: an AR model whose coefficients depend on whether q_{t-1} is below or above the threshold γ
  - $y_t = (\phi_{1,0} + \phi_1' \mathbf{y}_{t-1}) 1\{q_{t-1} \leq \gamma\} + (\phi_{2,0} + \phi_2' \mathbf{y}_{t-1}) 1\{q_{t-1} > \gamma\} + \varepsilon_t$
  - $\mathbf{y}_{t-1} = (y_{t-1}, \ldots, y_{t-p})'$: the last p values; $\phi_{j,0}$, ϕ_j: the intercept and AR coefficients of regime j
  - q_{t-1}: the observed threshold variable; γ: the threshold; $1\{\cdot\}$ selects the regime; variances may differ across regimes
- **SETAR** (self-exciting): $q_{t-1} = y_{t-d}$, the series itself with the **delay** lag d
  - notation SETAR($k; p_1, \ldots, p_k$): k regimes with AR orders $p_1, \ldots, p_k$ Tong and Lim (1980)

Threshold and self-exciting threshold autoregression (2/2)

- Dynamics: the **skeleton** $y_t = F(y_{t-1})$, the model with the noise switched off
  - F: the regime-dependent conditional mean; the skeleton can converge to a point, a limit cycle or a chaotic attractor
- Stationarity is a property of the whole model, not of each regime
  - SETAR(2; 1, 1) is ergodic iff $\phi_1 < 1$, $\phi_2 < 1$ and $\phi_1\phi_2 < 1$: a regime may be explosive on its own (Seminar 2, A5)
- Survey of threshold models in economics: Hansen (2011)

Estimation and inference (1/2)

- Concentrated least squares: for each candidate γ (and d) run OLS in both regimes
  - $\hat{\gamma} = \arg\min_{\gamma} S_T(\gamma)$
  - $S_T(\gamma)$: the total SSR given γ; the search runs over the central 70% of the observed values of q
  - $\hat{\gamma}$ is super-consistent, $T(\hat{\gamma} - \gamma) = O_p(1)$, with a nonstandard limit Chan (1993); the slopes are asymptotically Normal as if γ were known
- Testing linearity, H_0: $\phi_1 = \phi_2$: γ is not identified under H_0 (the Davies problem again) Hansen (1996)
  - $\sup_{\gamma} W_T(\gamma)$, with $W_T(\gamma)$ the Wald statistic of $\phi_1 = \phi_2$ at threshold γ, has a limit that depends on the data
  - **fixed-regressor bootstrap**: $y_t^* = \hat{e}_t \eta_t$, $\eta_t \sim N(0,1)$, same regressors, recompute sup W^*; $\hat{e}_t$: the residuals

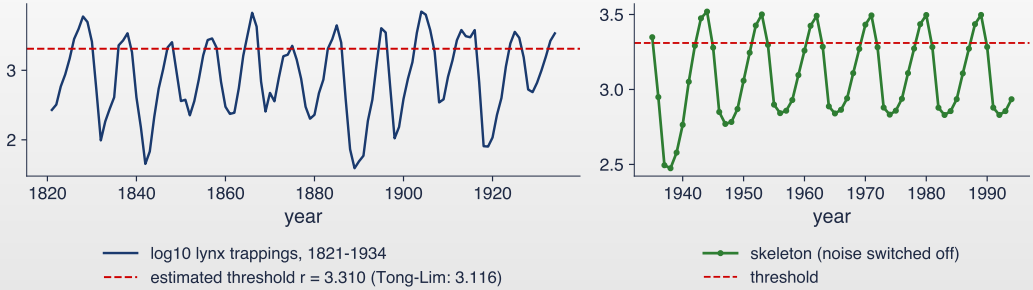
Estimation and inference (2/2)

- Confidence set for the threshold Hansen (2000): all γ with a small likelihood ratio
  - $\mathrm{LR}_T(\gamma) = T\,\dfrac{S_T(\gamma) - S_T(\hat{\gamma})}{S_T(\hat{\gamma})}$
  - the relative increase of the SSR when the threshold is moved from $\hat{\gamma}$ to γ
- Limit distribution and critical value
  - $\xi = \max_s[2W(s) - |s|]$, W a two-sided Brownian motion; $P(\xi \le x) = (1 - e^{-x/2})^2$
  - 95% value $-2\ln(1 - \sqrt{0.95}) = 7.35$: the 95% set is $\{\gamma : \mathrm{LR}_T(\gamma) \le 7.35\}$
  - under heteroskedasticity divide LR by $\hat{\eta}^2$, a variance-ratio correction estimated from the residuals (Appendix)

Case study: Tong and Lim (1980), the Canadian lynx

- The paper, Section 9: $\log_{10}$ of the yearly lynx trappings, 1821–1934; SETAR(2; 7, 2) with $d = 2$, threshold 3.116 Tong and Lim (1980)
  - $d = 2$ has an ecological reading: a lynx is fully grown in the autumn of its second year
- Our replication with the threshold of the paper: lower regime $y_t = 0.546 + 1.032y_{t-1} - 0.173y_{t-2} + \dots$ (seven lags), upper regime $2.345 + 1.533y_{t-1} - 1.276y_{t-2}$
  - the lower regime reproduces the published coefficients to three decimals; the free least-squares threshold is 3.310 (SSR 3.764 against 3.943)
- Question: does the fitted model generate the lynx cycle by itself?

The lynx cycle and the SETAR skeleton



- Left: data and the estimated threshold; right: the skeleton iterated 60 years from the last observations, noise switched off

ATS_ch2_lynx

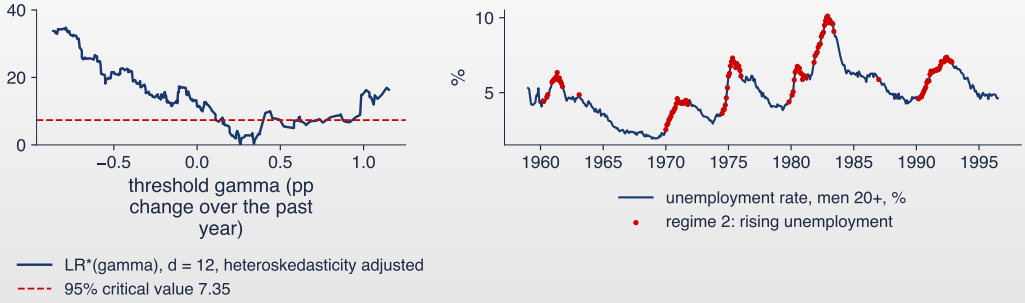
Interpreting the lynx SETAR

- The skeleton settles on a stable limit cycle of period 9 years: the model produces the cycle without any noise
  - a linear AR(11) needs noise to sustain the oscillation; its skeleton converges to the mean
- The rise is slow (lower regime, seven lags), the fall is fast (upper regime, an explosive AR(2) on its own): the asymmetry of the data
- Residual variance 0.0352 against 0.0364 for AR(11): a small in-sample gain, a large gain in the description of the dynamics

Case study: Hansen (1997), US unemployment

- The paper, Section 5: unemployment rate of men aged 20 and over (unemployed / labour force), monthly 1959.1–1996.7; Δy_t on a constant and 12 lags Hansen (1997)
  - y_t: the unemployment rate in %; threshold variable $q_{t-1} = y_{t-1} - y_{t-d}$, $d = 2, \ldots, 12$: the change over the last $d-1$ months; trimming 15%
  - robust sup-Wald test, bootstrap p-values (1000 replications); the LS estimate $\hat{d} = 12$, $\hat{\gamma} = 0.302$, 95% interval [0.213; 0.340], 314 and 124 observations
- Our replication: the same construction from the BLS series on FRED (today's seasonal adjustment), the same model, $d = 12$
- Question: is there a "recession regime" with different dynamics, and how precisely is its threshold estimated?

A threshold model for US unemployment



- Left: heteroskedasticity-adjusted $\mathrm{LR}^*(\gamma)$, $d = 12$; the 95% set is where the curve lies below 7.35. Right: months in the rising-unemployment regime ($q_{t-1} > \hat{\gamma}$)

ATS_ch2_unemp_tar

Interpreting the unemployment TAR

- $\hat{\gamma} = 0.261$ pp (paper: 0.302), regimes of 314 and 124 months, the same split as in Hansen (1997); 95% set [0.131, 0.917]
  - sup-Wald 47.2, bootstrap p 0.015; linearity is rejected at 5% for 10 of the 11 delays
- Rising regime: intercept 0.065, AR(1) 0.18, AR(2) 0.28; other months: -0.019, -0.15, 0.06
  - increases feed on themselves; in expansions unemployment is close to a random walk with a slight downward drift
- With today's data the SSR is smallest at $d = 3$ ($\hat{\gamma} = -0.24$): the delay is fragile, the threshold effect is not; 1959–2019: $\hat{\gamma} = 0.261$, $p < 0.001$

Recap: Threshold autoregression

- A piecewise-linear AR switching on an observed variable; can generate limit cycles
- $\hat{\gamma}$ super-consistent; slopes as if γ were known
- Linearity tests need the bootstrap; threshold intervals invert LR with the value 7.35
- Lynx and unemployment: both landmark results replicate

STAR models (1/2)

- A weighted average of two AR regimes, with a weight G that moves smoothly between 0 and 1 Teräsvirta (1994)
  - $y_t = \phi_1' x_t[1 - G(s_t; \gamma, c)] + \phi_2' x_t G(s_t; \gamma, c) + \varepsilon_t, \quad x_t = (1, y_{t-1}, \ldots, y_{t-p})'$
  - s_t: the transition variable (e.g. a lag of y_t); c: the location of the transition; $\gamma > 0$: its speed
  - $G = 0$: regime 1 with coefficients ϕ_1; $G = 1$: regime 2 with ϕ_2; in between, a mixture
- A continuum of regimes
  - smooth aggregation of many agents with different thresholds, or time aggregation, produces smooth switching

STAR models (2/2)

- **LSTAR** (logistic): low against high values of s_t
  - $G = [1 + \exp\{-\gamma(s_t - c)/\hat{\sigma}_s\}]^{-1}$
  - $\gamma \to \infty$ gives the TAR, $\gamma \to 0$ the linear AR; $\hat{\sigma}_s$: the standard deviation of s_t, which makes γ scale free and easier to estimate
- **ESTAR** (exponential): small against large *deviations* from c, symmetric
  - $G = 1 - \exp\{-\gamma(s_t - c)^2\}$
- Survey with the modelling cycle (specification, estimation, evaluation): van Dijk, Teräsvirta and Franses (2002)

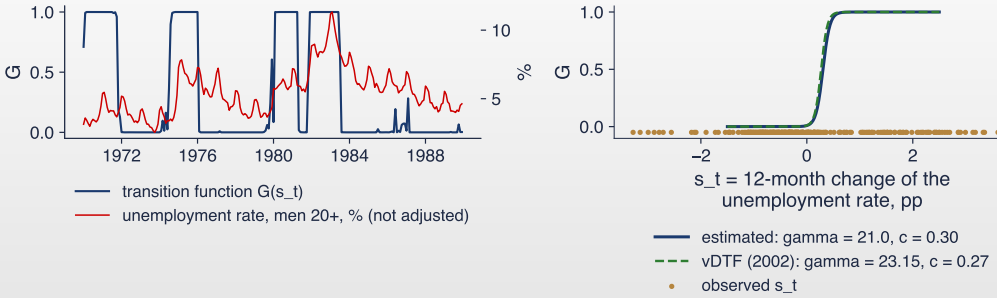
Testing linearity against STAR

- Under H_0: $\gamma = 0$, the parameters c and ϕ_2 are not identified: replace G by a third-order Taylor expansion around $\gamma = 0$ Luukkonen, Saikkonen and Teräsvirta (1988)
  - auxiliary regression $\hat{\varepsilon}_t = \beta_0' \mathrm{x}_t + \beta_1' \tilde{\mathrm{x}}_t s_t + \beta_2' \tilde{\mathrm{x}}_t s_t^2 + \beta_3' \tilde{\mathrm{x}}_t s_t^3 + v_t$
  - $\hat{\varepsilon}_t$: the residuals of the linear AR; $\tilde{\mathrm{x}}_t$: x_t without the constant; β_k: the coefficients of the interactions with s_t^k; v_t: the error
  - LM3: $\beta_1 = \beta_2 = \beta_3 = 0$, an F test with $3p$ restrictions; repeat for each candidate s_t and take the smallest p-value (derivation in the Appendix)
- **Choice of the family** Teräsvirta (1994): test H_{04}: $\beta_3 = 0$, then H_{03}: $\beta_2 = 0 \mid \beta_3 = 0$, then H_{02}: $\beta_1 = 0 \mid \beta_2 = \beta_3 = 0$
  - strongest rejection of H_{03}: ESTAR; of H_{04} or H_{02}: LSTAR (an ESTAR has no cubic term in its expansion)
- Estimation by nonlinear least squares, linear parameters concentrated out; evaluation by LM tests of no remaining nonlinearity and parameter constancy Eitrheim and Teräsvirta (1996)

Case study: van Dijk, Teräsvirta and Franses (2002)

- The paper, Section 7: unadjusted unemployment rate of US men aged 20+, June 1968–December 1999; estimation to 1989, forecasts for 1990–1999 van Dijk, Teräsvirta and Franses (2002)
  - Δy_t on a constant, 11 monthly dummies, y_{t-1} and 15 lags of Δy_t; transition variable $s_t = \Delta_{12} y_{t-d}$ (the 12-month change, lagged d months), $d = 1, \ldots, 6$
  - their LSTAR with $d = 1$: $\hat{\gamma} = 23.15$, $\hat{c} = 0.27$, residual s.d. 0.92 of the AR
- Our replication: the same series (BLS via FRED), sample, regressors and transition; the unrestricted lags, without their pruning of insignificant terms
- Question: does an LSTAR beat the linear model out of sample?

An LSTAR for US unemployment



- Left: the estimated transition function over time with the unemployment rate; right: G as a function of $s_t = \Delta_{12} y_{t-1}$, ours and the published one

ATS_ch2_unemp_lstar

Interpreting the LSTAR

- LM3 p-values: $d = 1$: 0.124, $d = 2$: 0.052, $d = 3$: 0.275; for $d = 2$: H_{04} 0.008, H_{03} 0.548, H_{02} 0.461: LSTAR
  - in line with van Dijk, Teräsvirta and Franses (2002): weak evidence, strongest for $d = 2$, pointing to the logistic family
- $\hat{\gamma} = 21.0$, $\hat{c} = 0.30$, residual s.d. ratio 0.90; the regime switches when unemployment has risen by about 0.3 pp in a year ($G > 0.5$ in 32% of months)
  - AIC prefers the LSTAR (-3.034 against -2.979); BIC the linear model (-2.573 against -2.366): 18 extra parameters without pruning
- 1990–1999, one step: RMSE 0.218 (LSTAR) against 0.228 (AR), HLN 1.14, p 0.255: better, not significantly

Real exchange rates and purchasing power parity



Gustav Cassel (1866–1945)

Portrait: unknown author; public domain; Wikimedia Commons



Bretton Woods, July 1944

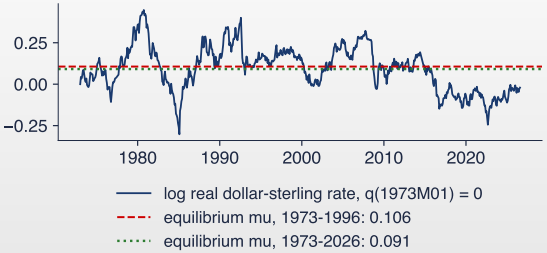
Photo: US National Archives (1944); public domain; Wikimedia Commons

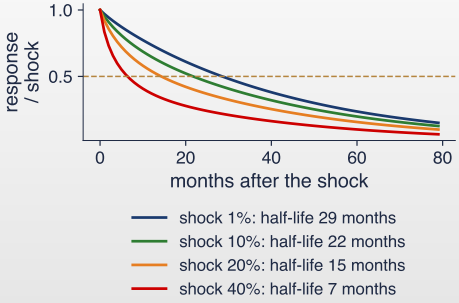
- Purchasing power parity, PPP (Cassel (1918)): the real exchange rate $q_t = s_t - p_t + p_t^*$ should be stationary
  - s_t: the log nominal rate (domestic per foreign currency); p_t, p_t^*: log domestic and foreign price levels
  - after Bretton Woods ended (1973) unit-root tests rarely reject
  - first PPP puzzle: no mean reversion; second: half-lives of 3–5 years, too slow for nominal shocks
- Transaction costs create a band of inaction: near parity q_t is close to a random walk, far from it arbitrage pulls it back Michael, Nobay and Peel (1997)
  - Taylor, Peel and Sarno (2001): ESTAR $q_t - \mu = (q_{t-1} - \mu)\exp\{-\theta^2(q_{t-1} - \mu)^2\} + \varepsilon_t$
  - μ: the equilibrium; near it the AR coefficient $\exp\{\cdot\}$ is close to 1 (random walk), far from it close to 0 (fast reversion); θ^2: the speed

Case study: Taylor, Peel and Sarno (2001)

- The paper: monthly real dollar rates of sterling, mark, franc and yen, 1973M01–1996M12, IMF IFS data, $q(1973M01) = 0$; ESTAR with $p = d = 1$ and $\beta_1 = -\beta_1^* = 1$ (eq. 8) Taylor, Peel and Sarno (2001)
  - dollar–sterling: $\hat{\theta}^2 = 0.452$, $\hat{\mu} = -0.149$, $s = 0.033$ (residual standard deviation); Monte Carlo p-value of $\hat{\theta}$ under a random walk: 0.002
  - half-lives from generalised impulse responses conditional on the average history: under one year for a 40% shock, just under three years for 1%
- Our replication for dollar–sterling: end-of-month DEXUSUK, US CPI (CPIAUCNS), UK CPI (OECD to 1987, then ONS); same model, same half-life procedure; extension to August 2026
- Question: is the real exchange rate nonlinearly mean reverting, and how fast?

Nonlinear mean reversion of the real dollar–sterling rate





⊡ Left: q_t and the estimated equilibria; right: generalised impulse responses of the 1973–1996 ESTAR, shocks of 1–40%, averaged over the observed histories

ATS_ch2_ppp_estar

Interpreting the ESTAR

- 1973–1996: $\hat{\theta}^{\mathbf{2}} = 0.527$ (s.e. 0.219), $\hat{\mu} = 0.106$, $s = 0.033$; Monte Carlo p-value of the t-ratio 0.165 (paper: 0.002)
  - $\theta^{\mathbf{2}}$ and s are close to the paper; μ differs because the UK CPI of the paper (IFS) is not the series published today
- Half-lives: 1% shock 29 months, 10% 22, 40% 7; linear AR(1): 21 months; from equilibrium: 42 months for 1%
  - large deviations die out fast: the second PPP puzzle is a feature of small shocks
- Dickey–Fuller: -2.20 (p 0.205); its power against our ESTAR is 19%
- KSS Kapetanios, Shin and Snell (2003): -2.47 (p 0.144) to 1996, -3.47 (p 0.014) to 2026
  - the t statistic of δ in $\Delta q_t = \delta q_{t-\mathbf{1}}^{\mathbf{3}} + e_t$ (demeaned q_t): a unit-root test against a stationary ESTAR; reject in the left tail

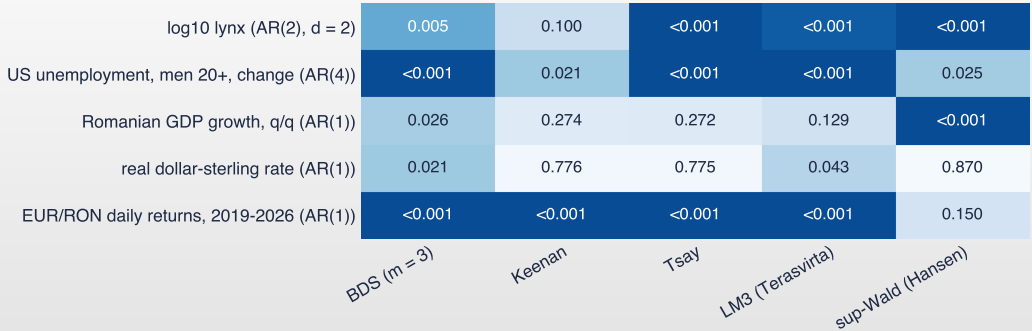
Recap: Smooth transition

- LSTAR: asymmetry between low and high states; ESTAR: small and large deviations
- LM tests from a Taylor expansion avoid the identification problem; the sequence H_{04}, H_{03}, H_{02} chooses the family
- Unemployment: the published LSTAR replicates; its forecast gain is small
- PPP: half-lives depend on the size of the shock; the evidence for nonlinearity depends on the sample

General and specific tests

- **BDS** Brock et al. (1996): correlation integral $C_m(\varepsilon)$ of m-histories; under i.i.d. $C_m = C_1^m$; applied to the residuals of a linear model
  - $C_m(\varepsilon)$: the share of pairs of m-histories $(y_t, \ldots, y_{t+m-1})$ that are within distance ε in every coordinate
  - power against any dependence left in the residuals, including ARCH: a rejection does not say which nonlinearity
- **Keenan** (squared fitted values) and the test of Tsay (1986) (all cross products $y_{t-i}y_{t-j}$, orthogonalised): F tests of second-order Volterra terms
  - Tsay's arranged-autoregression test targets the TAR alternative Tsay (1989)
- **LM3** (STAR) and **sup-Wald** (TAR): specific alternatives, more power when they are right
  - caution: structural breaks, outliers and heteroskedasticity also reject linearity Carrasco (2002)

Nonlinearity tests on five series



- p-values; darker: stronger rejection of linearity. BDS on the AR residuals ($m = 3$, $\varepsilon = \hat{\sigma}$); LM3 and sup-Wald with $s = q = y_{t-d}$; sup-Wald with 200 bootstrap replications

ATS_ch2_nonlinearity_tests

Interpreting the test battery

- Lynx and unemployment: every specific test rejects; the nonlinearity is real and of the threshold type
- EUR/RON returns: BDS, Keenan, Tsay and LM3 all reject with $p < 0.001$, yet the TAR sup-Wald does not (p 0.150)
  - volatility clustering and outliers, not a nonlinear conditional mean: the heteroskedasticity-robust test is the honest one
- Romanian GDP growth: only BDS and sup-Wald reject ($p < 0.001$): the 2009 and 2020 quarters act as a "regime"; a break or an outlier model may explain the same data

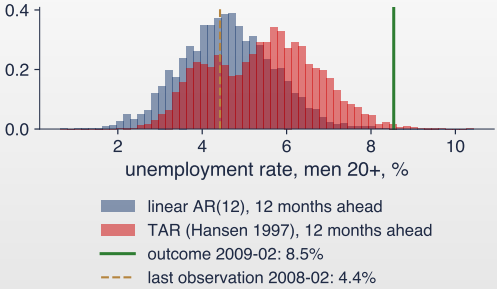
Threshold cointegration

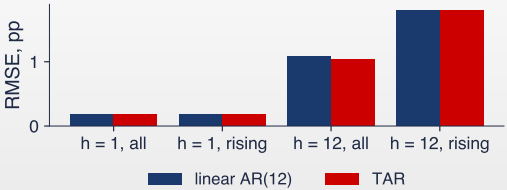
- Balke and Fomby (1997): the error-correction term $z_t = y_t - \beta x_t$ adjusts only outside a band: $\Delta z_t = \rho\, z_{t-1} 1\{|z_{t-1}| > \gamma\} + \varepsilon_t$
  - y_t, x_t: two cointegrated prices; β: the cointegrating coefficient; $\rho < 0$: the speed of adjustment outside the band $[-\gamma, \gamma]$
  - the same transaction-cost argument as ESTAR, for a pair of prices: commodity prices in two markets, interest-rate pass-through, the law of one price
- Asymmetric adjustment (momentum TAR) Enders and Siklos (2001); the test of linear against threshold VECM with a bootstrap Hansen and Seo (2002)
  - Chapter 4 introduces the linear VECM and ARDL; threshold versions are a natural project extension
- Romanian example: pass-through from the BNR policy rate to ROBOR and to lending rates, with a band where banks do not adjust

Multi-step forecasts and the model skeleton

- For $h = 1$: $\mathbb{E}(y_{t+1} \mid \mathcal{F}_t) = F(\mathrm{y}_t)$; for $h \geq 2$: $\mathbb{E}[F(F(\mathrm{y}_t) + \varepsilon_{t+1}, \dots)] \neq F(F(\mathrm{y}_t))$ (Jensen)
  - F: the conditional mean function of the model; Jensen: the mean of a nonlinear function is not the function of the mean
  - iterating the skeleton (the "naive" forecast) is biased; simulate paths with bootstrapped residuals and average them Clements, Franses and Swanson (2004)
  - the predictive density can be skewed or bimodal: report densities, scored by the CRPS or the log score (Chapter 1)
- Evidence: nonlinear models rarely beat linear ones on average; gains concentrate in particular regimes (recessions) Montgomery et al. (1998), Teräsvirta, van Dijk and Medeiros (2005)
  - evaluate conditionally: GW tests with the regime as instrument (Chapter 1); asymmetric dynamics in US unemployment Koop and Potter (1999)

TAR and AR forecasts of US unemployment





- Hansen's TAR ($d = 12$) and a linear AR(12) in differences, parameters fixed at 1959–1996; forecasts of the level by 500 simulated paths, origins 1996–2019. Left: 12-month densities from February 2008

ATS_ch2_nonlinear_forecasts

Interpreting the nonlinear forecasts

- From February 2008 (before the 2008–2009 surge): TAR mean 5.5%, 90% range [3.5; 7.5]; AR mean 4.6%, [2.9; 6.4]; outcome 8.5%
  - the TAR density is skewed upwards once unemployment starts rising; neither model foresaw the size of the recession
- $P = 270$ months: RMSE at 1 month 0.173 (TAR) against 0.177 (AR), HLN 1.12 (p 0.265); at 12 months 1.039 against 1.087 (p 0.599)
  - in the 62 rising-regime months the two are equal at 12 months (1.802 and 1.801): no regime-specific gain out of sample
- A good in-sample nonlinearity is not a forecasting gain: pre-register the comparison

Recap: Nonlinearity tests and forecasts

- General tests (BDS, Keenan, Tsay) detect, specific ones (LM3, sup-Wald) describe
- Heteroskedasticity, outliers and breaks also reject linearity
- Multi-step nonlinear forecasts need simulation; report densities
- Out of sample, nonlinear models gain little on average; test where they should gain

An open question

- Nonlinearity or breaks? A few shifts of the mean can look like nonlinear mean reversion, and a threshold model can look like a sequence of breaks Carrasco (2002), Diebold and Inoue (2001)
  - formal: is the KSS (ESTAR) rejection for the real dollar–sterling rate robust to Bai–Perron mean regimes estimated on the same data?
  - falsified if the rejection disappears once the null distribution accounts for the estimated breaks
- Why it matters: half-lives, PPP and the credibility of exchange-rate models depend on which story is true
  - literature to start from: Taylor, Peel and Sarno (2001), Kapetanios, Shin and Snell (2003), Michael, Nobay and Peel (1997), Carrasco (2002), Bai and Perron (2003)

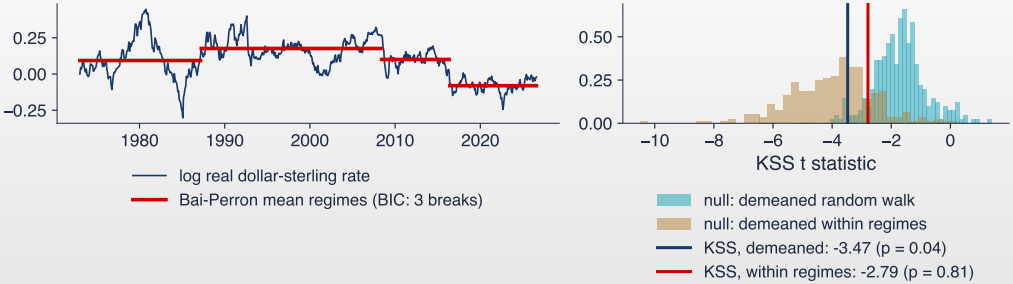
The discovery loop with an AI assistant

- An AI assistant (an LLM such as Claude, ChatGPT, Gemini or Copilot) speeds up each step; Semantic Scholar and Elicit help with the literature
  - **literature**: `List peer-reviewed papers that test ESTAR real exchange rate models allowing for breaks in the mean; give DOIs.` Then check every DOI on Crossref
  - **hypothesis**: `Propose a data-generating process with mean breaks and no nonlinearity that makes the KSS test reject.`
  - **code and replication**: reproduce first a number of this lecture (the KSS statistic to 2026), then the new procedure
  - **robustness and critique**: `Act as a referee: list every way in which estimating breaks on the same data biases the test.`
- Report: what was asked, what was kept, what was rejected (AI_USE.md, AI_ERRORS.md)

Required checks

- Every reference exists and says what is claimed (DOI resolves, title matches, the result is in the paper)
- The null distribution reproduces the whole procedure: breaks estimated on each simulated series, not fixed at the observed dates
- The number of breaks, the trimming and the criterion (BIC) are fixed before seeing the test result
- The data are the published series (vintage, deflator, end of month); differences from the paper are reported
- "Not significant within regimes" is not "linear": report the power of the procedure

Mini-case: the KSS test after Bai–Perron mean shifts



- Left: real dollar–sterling rate with Bai–Perron mean regimes (BIC: three breaks, March 1987, May 2008, May 2016); right: KSS null distributions from 300 random walks put through the same steps
- Demeaned: KSS -3.47, p 0.037 (5%: -3.13); within regimes: -2.79, p 0.813 (5%: -6.30): the evidence for ESTAR disappears once the break search is part of the null

ATS_ch2_ai_case

Project idea

- **Regimes in Romanian inflation and the policy rate**: breaks, thresholds or both?
  - replicate first: the Bai–Perron dates of this lecture and the TAR test of Hansen (1997) on the US data
  - extension: an AR model of Romanian inflation with a partial break in the intercept against an LSTAR in the 12-month change of inflation; pre-register the comparison (1–12 months, CRPS, Holm (1979))
  - optional: threshold pass-through from the BNR rate to ROBOR (Chapter 4)
- Deliverables follow the course rules: repository, report, AI_USE.md, AI_ERRORS.md, oral defence

Key takeaways

- A break date chosen from the data is a nuisance parameter: use sup/exp/ave tests with their own critical values
- Bai–Perron: global LS by dynamic programming, UDmax and sequential tests, BIC/LWZ, intervals for the dates
- Monitor with a CSW boundary; test variance breaks with κ_2; bootstrap unit-root tests with breaks
- TAR and STAR: test with the bootstrap or LM expansions, estimate by concentrated (N)LS, forecast by simulation
- Breaks and nonlinearity mimic each other: test one allowing for the other

Self-assessment

Questions

- Why is the 5% critical value of sup-Wald with one coefficient above 8, not 3.84?
- What does dynamic programming save in the Bai–Perron estimator?
- Why does a repeated 5% test eventually raise a false alarm?
- Why is the threshold of a TAR estimated at rate T?
- When does the half-life of an ESTAR depend on the shock?

Next: SVAR and local projections

- Structural VAR and local projections: identification of shocks
- State-dependent local projections extend this chapter's nonlinear models to impulse responses

Appendix: the limit of the sup-Wald statistic

- Mean shift, σ^2 known: $W_T(\pi) = \dfrac{T\pi(1-\pi)(\bar{y}_1 - \bar{y}_2)^2}{\sigma^2}$, $\bar{y}_1$, $\bar{y}_2$ the means before and after $[\pi T]$
- With $S_T(r) = \sigma^{-1} T^{-1/2} \sum_{t \le [rT]} (y_t - \mu) \Rightarrow B(r)$ (FCLT, Chapter 0):
  $\bar{y}_1 - \bar{y}_2 = \dfrac{\sigma}{\sqrt{T}} \left[\dfrac{S_T(\pi)}{\pi} - \dfrac{S_T(1) - S_T(\pi)}{1-\pi} \right]$
- The bracket equals $\dfrac{S_T(\pi) - \pi S_T(1)}{\pi(1-\pi)}$, so $W_T(\pi) \Rightarrow \dfrac{[B(\pi) - \pi B(1)]^2}{\pi(1-\pi)}$, a squared Brownian bridge over its variance
- The continuous mapping theorem gives the limit of $\sup_{\pi \in \Pi}$; with p coefficients, B is p-dimensional; with an unknown σ^2 or a HAC variance the limit is the same

Back

Appendix: the critical value 7.35 of the threshold LR

- Hansen (2000): with a shrinking threshold effect, $\mathrm{LR}_T(\gamma_0) \to_d \eta^2 \xi$, $\xi = \max_s[2W(s) - |s|]$, W a two-sided Brownian motion
- For one side, $\max_{s \geq 0}[2W(s) - s]$ is exponential with mean 2: $P(\cdot \leq x) = 1 - e^{-x/2}$ (the maximum of a Brownian motion with drift)
- The two sides are independent: $P(\xi \leq x) = (1 - e^{-x/2})^2$; solving $(1 - e^{-x/2})^2 = 1 - \alpha$ gives $c(\alpha) = -2\ln(1 - \sqrt{1 - \alpha})$
- $c(0.10) = 5.94$, $c(0.05) = 7.35$, $c(0.01) = 10.59$; under heteroskedasticity divide LR by $\hat{\eta}^2$ before comparing

Back

Appendix: the Taylor expansion behind LM3

- LSTAR: $G(s;\gamma,c) - \frac{1}{2} = \frac{1}{4}\gamma(s-c) - \frac{1}{48}\gamma^3(s-c)^3 + O(\gamma^5)$ around $\gamma = 0$ (with $\hat{\sigma}_s = 1$)
- Substituting in $\phi_1' x_t + (\phi_2 - \phi_1)' x_t G$ and collecting powers of s_t gives $\beta_0' x_t + \sum_{k=1}^{3} \beta_k' \tilde{x}_t s_t^k$, every β_k proportional to γ
- So H_0: $\gamma = 0$ becomes $\beta_1 = \beta_2 = \beta_3 = 0$, testable without estimating c; the first-order expansion (LM1) has no power when only the intercept switches Luukkonen, Saikkonen and Teräsvirta (1988)
- ESTAR: $1 - e^{-\gamma(s-c)^2} = \gamma(s-c)^2 + O(\gamma^2)$: squares only, hence the decision rule of Teräsvirta (1994)

Back

References (1/7)

- Andrews, D. W. K. (1993). Tests for parameter instability and structural change with unknown change point. *Econometrica*, 61(4), 821–856. doi:10.2307/2951764
- Andrews, D. W. K., & Ploberger, W. (1994). Optimal tests when a nuisance parameter is present only under the alternative. *Econometrica*, 62(6), 1383–1414. doi:10.2307/2951753
- Bai, J. (1997). Estimation of a change point in multiple regression models. *The Review of Economics and Statistics*, 79(4), 551–563. doi:10.1162/003465397557132
- Bai, J., & Perron, P. (1998). Estimating and testing linear models with multiple structural changes. *Econometrica*, 66(1), 47–78. doi:10.2307/2998540
- Bai, J., & Perron, P. (2003). Computation and analysis of multiple structural change models. *Journal of Applied Econometrics*, 18(1), 1–22. doi:10.1002/jae.659
- Balke, N. S., & Fomby, T. B. (1997). Threshold cointegration. *International Economic Review*, 38(3), 627–645. doi:10.2307/2527284
- Brock, W. A., Dechert, W. D., Scheinkman, J. A., & LeBaron, B. (1996). A test for independence based on the correlation dimension. *Econometric Reviews*, 15(3), 197–235. doi:10.1080/07474939608800353
- Brown, R. L., Durbin, J., & Evans, J. M. (1975). Techniques for testing the constancy of regression relationships over time. *Journal of the Royal Statistical Society, Series B*, 37(2), 149–192. doi:10.1111/j.2517-6161.1975.tb01532.x
- Carrasco, M. (2002). Misspecified structural change, threshold, and Markov-switching models. *Journal of Econometrics*, 109(2), 239–273. doi:10.1016/S0304-4076(02)00112-4
- Casini, A., & Perron, P. (2019). Structural breaks in time series. In *Oxford Research Encyclopedia of Economics and Finance*. Oxford University Press. doi:10.1093/acrefore/9780190625979.013.179
- Cassel, G. (1918). Abnormal deviations in international exchanges. *The Economic Journal*, 28(112), 413–415. doi:10.2307/2223329

References (2/7)

- Chan, K. S. (1993). Consistency and limiting distribution of the least squares estimator of a threshold autoregressive model. *The Annals of Statistics*, 21(1), 520–533. doi:10.1214/aos/1176349040
- Chow, G. C. (1960). Tests of equality between sets of coefficients in two linear regressions. *Econometrica*, 28(3), 591–605. doi:10.2307/1910133
- Chu, C.-S. J., Hornik, K., & Kuan, C.-M. (1995). MOSUM tests for parameter constancy. *Biometrika*, 82(3), 603–617. doi:10.1093/biomet/82.3.603
- Chu, C.-S. J., Stinchcombe, M., & White, H. (1996). Monitoring structural change. *Econometrica*, 64(5), 1045–1065. doi:10.2307/2171955
- Clements, M. P., Franses, P. H., & Swanson, N. R. (2004). Forecasting economic and financial time-series with non-linear models. *International Journal of Forecasting*, 20(2), 169–183. doi:10.1016/j.ijforecast.2003.10.004
- Clements, M. P., & Hendry, D. F. (1998). *Forecasting Economic Time Series*. Cambridge University Press. doi:10.1017/CBO9780511599286
- Clements, M. P., & Hendry, D. F. (2006). Forecasting with breaks. In G. Elliott, C. W. J. Granger, & A. Timmermann (Eds.), *Handbook of Economic Forecasting* (Vol. 1, pp. 605–657). Elsevier. doi:10.1016/S1574-0706(05)01012-8
- Davies, R. B. (1987). Hypothesis testing when a nuisance parameter is present only under the alternative. *Biometrika*, 74(1), 33–43. doi:10.1093/biomet/74.1.33
- Diebold, F. X., & Inoue, A. (2001). Long memory and regime switching. *Journal of Econometrics*, 105(1), 131–159. doi:10.1016/S0304-4076(01)00073-2
- Diebold, F. X., & Mariano, R. S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263. doi:10.1080/07350015.1995.10524599
- van Dijk, D., Teräsvirta, T., & Franses, P. H. (2002). Smooth transition autoregressive models: A survey of recent developments. *Econometric Reviews*, 21(1), 1–47. doi:10.1081/ETC-120008723

References (3/7)

- Eitrheim, Ø., & Teräsvirta, T. (1996). Testing the adequacy of smooth transition autoregressive models. *Journal of Econometrics*, 74(1), 59–75. doi:10.1016/0304-4076(95)01751-8
- Enders, W., & Siklos, P. L. (2001). Cointegration and threshold adjustment. *Journal of Business & Economic Statistics*, 19(2), 166–176. doi:10.1198/07350010131697039 5
- Fryzlewicz, P. (2014). Wild binary segmentation for multiple change-point detection. *The Annals of Statistics*, 42(6), 2243–2281. doi:10.1214/14-AOS1245
- Garcia, R., & Perron, P. (1996). An analysis of the real interest rate under regime shifts. *The Review of Economics and Statistics*, 78(1), 111–125. doi:10.2307/2109851
- Giacomini, R., & Rossi, B. (2009). Detecting and predicting forecast breakdowns. *The Review of Economic Studies*, 76(2), 669–705. doi:10.1111/j.1467-937X.2009.00545.x
- Giacomini, R., & Rossi, B. (2010). Forecast comparisons in unstable environments. *Journal of Applied Econometrics*, 25(4), 595–620. doi:10.1002/jae.1177
- Hamilton, J. D. (1994). *Time Series Analysis*. Princeton University Press. doi:10.2307/j.ctv14jx6sm
- Hansen, B. E. (1996). Inference when a nuisance parameter is not identified under the null hypothesis. *Econometrica*, 64(2), 413–430. doi:10.2307/2171789
- Hansen, B. E. (1997). Inference in TAR models. *Studies in Nonlinear Dynamics & Econometrics*, 2(1). doi:10.2202/1558-3708.1024
- Hansen, B. E. (2000). Sample splitting and threshold estimation. *Econometrica*, 68(3), 575–603. doi:10.1111/1468-0262.00124
- Hansen, B. E. (2001). The new econometrics of structural change: Dating breaks in U.S. labor productivity. *Journal of Economic Perspectives*, 15(4), 117–128. doi:10.1257/jep.15.4.117

References (4/7)

- Hansen, B. E. (2011). Threshold autoregression in economics. *Statistics and Its Interface*, 4(2), 123–127. doi:10.4310/SII.2011.v4.n2.a4
- Hansen, B. E., & Seo, B. (2002). Testing for two-regime threshold cointegration in vector error-correction models. *Journal of Econometrics*, 110(2), 293–318. doi:10.1016/S0304-4076(02)00097-0
- Harvey, D., Leybourne, S., & Newbold, P. (1997). Testing the equality of prediction mean squared errors. *International Journal of Forecasting*, 13(2), 281–291. doi:10.1016/S0169-2070(96)00719-4
- Holm, S. (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2), 65–70. www.jstor.org
- Huang, C., & Petukhina, A. (2022). *Applied Time Series Analysis and Forecasting with Python*. Springer. doi:10.1007/978-3-031-13584-2
- Inclán, C., & Tiao, G. C. (1994). Use of cumulative sums of squares for retrospective detection of changes of variance. *Journal of the American Statistical Association*, 89(427), 913–923. doi:10.1080/01621459.1994.10476824
- Inoue, A., Jin, L., & Rossi, B. (2017). Rolling window selection for out-of-sample forecasting with time-varying parameters. *Journal of Econometrics*, 196(1), 55–67. doi:10.1016/j.jeconom.2016.03.006
- Kapetanios, G., Shin, Y., & Snell, A. (2003). Testing for a unit root in the nonlinear STAR framework. *Journal of Econometrics*, 112(2), 359–379. doi:10.1016/S0304-4076(02)00202-6
- Kilian, L., & Lütkepohl, H. (2017). *Structural Vector Autoregressive Analysis*. Cambridge University Press. doi:10.1017/9781108164818
- Killick, R., Fearnhead, P., & Eckley, I. A. (2012). Optimal detection of changepoints with a linear computational cost. *Journal of the American Statistical Association*, 107(500), 1590–1598. doi:10.1080/01621459.2012.737745
- Kim, D., & Perron, P. (2009). Unit root tests allowing for a break in the trend function at an unknown time under both the null and alternative hypotheses. *Journal of Econometrics*, 148(1), 1–13. doi:10.1016/j.jeconom.2008.08.019

References (5/7)

- Koop, G., & Potter, S. M. (1999). Dynamic asymmetries in U.S. unemployment. *Journal of Business & Economic Statistics*, 17(3), 298–312. doi:10.1080/07350015.1999.10524819
- Lee, J., & Strazicich, M. C. (2003). Minimum Lagrange multiplier unit root test with two structural breaks. *The Review of Economics and Statistics*, 85(4), 1082–1089. doi:10.1162/003465303772815961
- Liu, J., Wu, S., & Zidek, J. V. (1997). On segmented multivariate regression. *Statistica Sinica*, 7(2), 497–525. www3.stat.sinica.edu.tw
- Lumsdaine, R. L., & Papell, D. H. (1997). Multiple trend breaks and the unit-root hypothesis. *The Review of Economics and Statistics*, 79(2), 212–218. doi:10.1162/003465397556791
- Luukkonen, R., Saikkonen, P., & Teräsvirta, T. (1988). Testing linearity against smooth transition autoregressive models. *Biometrika*, 75(3), 491–499. doi:10.1093/biomet/75.3.491
- McConnell, M. M., & Perez-Quiros, G. (2000). Output fluctuations in the United States: What has changed since the early 1980's? *American Economic Review*, 90(5), 1464–1476. doi:10.1257/aer.90.5.1464
- Michael, P., Nobay, A. R., & Peel, D. A. (1997). Transactions costs and nonlinear adjustment in real exchange rates: An empirical investigation. *Journal of Political Economy*, 105(4), 862–879. doi:10.1086/262096
- Montgomery, A. L., Zarnowitz, V., Tsay, R. S., & Tiao, G. C. (1998). Forecasting the U.S. unemployment rate. *Journal of the American Statistical Association*, 93(442), 478–493. doi:10.1080/01621459.1998.10473696
- Neftçi, S. N. (1984). Are economic time series asymmetric over the business cycle? *Journal of Political Economy*, 92(2), 307–328. doi:10.1086/261226
- Oka, T., & Perron, P. (2018). Testing for common breaks in a multiple equations system. *Journal of Econometrics*, 204(1), 66–85. doi:10.1016/j.jeconom.2018.01.003
- Perron, P. (1989). The great crash, the oil price shock, and the unit root hypothesis. *Econometrica*, 57(6), 1361–1401. doi:10.2307/1913712

References (6/7)

- Pesaran, M. H., & Pick, A. (2011). Forecast combination across estimation windows. *Journal of Business & Economic Statistics*, 29(2), 307–318. doi:10.1198/jbes.2010.09018
- Pesaran, M. H., & Timmermann, A. (2007). Selection of estimation window in the presence of breaks. *Journal of Econometrics*, 137(1), 134–161. doi:10.1016/j.jeconom.2006.03.010
- Petropoulos, F., Apiletti, D., Assimakopoulos, V., et al. (2022). Forecasting: theory and practice. *International Journal of Forecasting*, 38(3), 705–871. doi:10.1016/j.ijforecast.2021.11.001
- Ploberger, W., & Krämer, W. (1992). The CUSUM test with OLS residuals. *Econometrica*, 60(2), 271–285. doi:10.2307/2951597
- Quandt, R. E. (1960). Tests of the hypothesis that a linear regression system obeys two separate regimes. *Journal of the American Statistical Association*, 55(290), 324–330. doi:10.1080/01621459.1960.10482067
- Robbins, H., & Siegmund, D. (1970). Boundary crossing probabilities for the Wiener process and sample sums. *The Annals of Mathematical Statistics*, 41(5), 1410–1429. doi:10.1214/aoms/1177696787
- Rossi, B. (2013). Advances in forecasting under instability. In G. Elliott & A. Timmermann (Eds.), *Handbook of Economic Forecasting* (Vol. 2B, pp. 1203–1324). Elsevier. doi:10.1016/B978-0-444-62731-5.00021-X
- Sansó, A., Aragó, V., & Carrion, J. L. (2004). Testing for changes in the unconditional variance of financial time series. *Revista de Economía Financiera*, 4, 32–52. econpapers.repec.org
- Stock, J. H., & Watson, M. W. (1996). Evidence on structural instability in macroeconomic time series relations. *Journal of Business & Economic Statistics*, 14(1), 11–30. doi:10.1080/07350015.1996.10524626
- Taylor, M. P., Peel, D. A., & Sarno, L. (2001). Nonlinear mean-reversion in real exchange rates: Toward a solution to the purchasing power parity puzzles. *International Economic Review*, 42(4), 1015–1042. doi:10.1111/1468-2354.00144
- Teräsvirta, T. (1994). Specification, estimation, and evaluation of smooth transition autoregressive models. *Journal of the American Statistical Association*, 89(425), 208–218. doi:10.1080/01621459.1994.10476462

References (7/7)

- Teräsvirta, T., van Dijk, D., & Medeiros, M. C. (2005). Linear models, smooth transition autoregressions, and neural networks for forecasting macroeconomic time series: A re-examination. *International Journal of Forecasting*, 21(4), 755–774. doi:10.1016/j.ijforecast.2005.04.010
- Tong, H., & Lim, K. S. (1980). Threshold autoregression, limit cycles and cyclical data. *Journal of the Royal Statistical Society, Series B*, 42(3), 245–268. doi:10.1111/j.2517-6161.1980.tb01126.x
- Truong, C., Oudre, L., & Vayatis, N. (2020). Selective review of offline change point detection methods. *Signal Processing*, 167, 107299. doi:10.1016/j.sigpro.2019.107299
- Tsay, R. S. (1986). Nonlinearity tests for time series. *Biometrika*, 73(2), 461–466. doi:10.1093/biomet/73.2.461
- Tsay, R. S. (1989). Testing and modeling threshold autoregressive processes. *Journal of the American Statistical Association*, 84(405), 231–240. doi:10.1080/01621459.1989.10478760
- Zeileis, A., Leisch, F., Hornik, K., & Kleiber, C. (2002). strucchange: An R package for testing for structural change in linear regression models. *Journal of Statistical Software*, 7(2), 1–38. doi:10.18637/jss.v007.i02
- Zivot, E., & Andrews, D. W. K. (1992). Further evidence on the great crash, the oil-price shock, and the unit-root hypothesis. *Journal of Business & Economic Statistics*, 10(3), 251–270. doi:10.1080/07350015.1992.10509904